Veeramallu Sai Abhiram

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Summary

Final-year B.Tech Computer Science student (CGPA: 9.23, graduating 2027) and Full Stack Developer intern with hands-on experience building and shipping full-stack applications using **Python, React, TypeScript, and SQL**. Experienced in frontend, integrating REST APIs, working in Agile teams, and building AI-powered features, backed by strong data structures and algorithms fundamentals and a habit of learning new tools quickly.

Education

Vellore Institute of Technology (VIT) — Vellore, Tamil Nadu
Bachelor of Technology (B.Tech) in Computer Science; CGPA: 9.23/10 — *July 2023 – May 2027*

Sri Chaitanya Junior College — Visakhapatnam, Andhra Pradesh
Intermediate (Class XII), MPC; 96% — *Jun 2021 – May 2023*

Experience

Full Stack Developer Intern [Certificate] — May 2026 – Jul 2026
Watania Solutions Pvt. Ltd. — *Hyderabad, Telangana*

- Developed and shipped production **React Native/TypeScript** screens for the WSC HR Portal, translating UI/UX specs into reusable components reviewed and approved by senior engineers.
- Integrated and **tested RESTful APIs** for authentication, employee management, attendance, and leave workflows; wrote test cases that caught data-flow and error-handling regressions before release.
- Resolved cross-team integration issues by validating API contracts with backend engineers and tracing request failures across the stack in a fast-paced **Scrum/Agile** environment.
- Collaborated with design and backend teams through sprint planning, code reviews, and daily stand-ups, clearly communicating technical trade-offs.

Projects

Unified Ride & Parcel Route Optimization Platform | GitHub | Live Demo — Jun 2026
Python, FastAPI, PostgreSQL, SQLAlchemy, React, TypeScript, JWT

- Solved a real-world **optimization problem**: implemented the **Hungarian (Kuhn-Munkres) algorithm from scratch** to assign captains to rides and bundle parcels, minimizing total cost using **Haversine-based distance** metrics.
- Built the **FastAPI** backend with **SQLAlchemy and PostgreSQL**, and a React/TypeScript front end, to dispatch combined passenger-ride and parcel requests across multiple captains.
- Prevented double-booking under simultaneous requests using **optimistic concurrency control**, version-checked atomic updates, and transactional acceptance.
- Implemented **JWT authentication** with access/refresh tokens and token revocation; deployed the app on **Vercel**.

AI Bot Detection Platform | GitHub | Live Demo — Apr 2026
Python (scikit-learn, XGBoost, SHAP), Node.js, Express, PostgreSQL, Docker, Selenium, Playwright, Puppeteer

- Built an end-to-end **machine learning** bot-detection platform (**Python, Node.js/Express, PostgreSQL, Docker**) that scores sessions using browser behavior, automation signals, and honeypots, combining **rule-based risk scoring** with **Logistic Regression and XGBoost**.
- Created a **105-session labeled dataset** from real Selenium, Playwright, and Puppeteer bots plus synthetic human sessions; engineered features, then trained, evaluated, and compared **Logistic Regression vs. XGBoost** models.
- Used **SHAP** for model explainability and **red-teamed** the detector with Puppeteer Stealth, measuring a **50% evasion rate** and identifying weaknesses in client-side detection.
- Integrated **Google Gemini** to turn detection results into structured threat reports using **versioned prompts** and evidence-based output checks.

Technical Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, C, C++, HTML5, CSS3
Front End: React, React Native, Responsive UI, Reusable Components
Back End & Databases: FastAPI, Node.js, Express, REST APIs, JWT, PostgreSQL, SQLite, SQLAlchemy
AI/ML: scikit-learn, XGBoost, Logistic Regression, SHAP, Google Gemini API (LLM integration)
Tools & Testing: Git, GitHub, Docker, Vercel, Postman, Agile/Scrum, API/Unit Testing, Selenium, Playwright
Core CS: DSA, OOP, Operating Systems, Computer Networks, DBMS, Software Engineering, Cloud Architecture

Certifications & Achievements

IBM Generative AI Engineering Professional Certificate [Certificate] — June 2025
VIT Merit Scholarship for Academic Excellence [Certificate] — 2023–2024